Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ from the Trace Structure of K_4 and Independence of Gravitational Engagement

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Abstract

The Projective Dynamic Logo (PDL) framework derives fundamental constants from four axioms on finite signed graphs. Paper D31 of the corpus (Gate 3) established that the effective gravitational coupling at nucleon number N satisfies $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ under the linear bridge inversion hypothesis. The present paper replaces that hypothesis with two explicitly named, independently motivated conditions and provides a rigorous proof of Gate 3 under them.

Four results are established. *Proposition 1* proves that the gravitational engagements of N distinct protons on the surface of K_4 are *exactly* statistically independent: the cross-signs of distinct protons belong to disjoint variable spaces, so the condition $(A) \wedge (B)$ on proton k places no constraint on proton $l \neq k$. The correlation $\delta = P(\text{joint}) - P(\text{marginal})^2$ vanishes exactly and is verified exhaustively over all 256 cross-sign pairs. *Lemma 1* (occupation) derives from Proposition 1 and the S_4-transitivity of D31 that $P(\text{edge } e \text{ active} \mid N \text{ protons}) = \sigma(N)$ uniformly for every edge e. *Lemma 2* (trace cancellation) proves that $\text{Tr}(A) = 0$ for any graph-adjacency matrix A, so that any S_4-invariant coupling $\Phi = \alpha I_6 + \beta A$ yields $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ regardless of β. *Theorem 1* (Gate 3 strengthened) assembles these results: under Hypothesis H1 ($\kappa = R_{\text{surf}}/R_{\text{tot}}$ is the gravitational engagement fraction of a single proton) and Hypothesis H2 (the bridge formula of D20), $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ for all $N \geq 0$, with the linear bridge inversion hypothesis of D31 superseded.

The document identifies the sole residual open problem OP1: deriving H1, i.e. justifying $\kappa = R_{\text{surf}}/R_{\text{tot}}$ as the gravitational engagement fraction of a single proton from axioms C1–C4, without invoking D25 as a definition. The document is self-contained: Table 1 (condition (B)), Table 2 (exhaustive cross-sign enumeration), and Table 3 (independence verification) allow the reader to reproduce all key steps independently.

1 Introduction

The PDL programme reconstructs fundamental physics from four axioms on finite signed graphs [8, 7]. The proton is characterised by the unique integer quintuplet
$(n_u, n_d, \mathrm{r}, \mathrm{R}_{\text{sea}}, \mathrm{R}_{\text{tot}}) = (24, 28, 930, 10087, 11017)$,
from which the fine-structure constant $\alpha_{\text{PDL}}^{-1} = \varphi^2 \mathrm{r}^2/(9\mu) = 137.022$ and the gravitational constant $G_{\text{PDL}} = 6.67448 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$ are derived without free parameter [4, 3].

Paper D25 introduced the surface coverage fraction $\sigma(N) = 1-(1-\kappa)^N$ with $\kappa = R_{\text{surf}}/R_{\text{tot}} \in \mathbb{Q}(\sqrt{5})$ as the relational density of an ensemble of N nucleons [5]. Paper D31 (Gate 3) proved $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ under the *linear bridge inversion hypothesis*, which stated that G scales linearly with $\sigma(N)$ [6]. The hypothesis was supported numerically to 0.004% (bridge residual of D20) and 0.006% (Hubble tension ratio), but its algebraic origin was not established. This constituted open problem OP1 of D31.

The present paper resolves OP1 by identifying the correct algebraic source of linearity and replacing the vague hypothesis with two precisely stated conditions. The key insight from the

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Colab diagnostics of Session 15 is twofold. First, a naive substitution $\varepsilon_G \mapsto \sigma(N) \cdot \varepsilon_G$ in the bridge formula $G = (\hbar c/m_p^2) \cdot \varepsilon_G^{18}$ yields $\sigma(N)^{18} \cdot G_{\mathrm{PDL}}$, which is numerically incompatible with Gate 3 by factors of order 10^{38} at $N = 40$. The linearity therefore does not arise from scaling the leakage parameter. Second, G is correctly understood as a trace functional over the active surface of K_4: since each proton contributes independently to the active surface (Proposition 1) and the S_4-invariant part of the coupling matrix has a vanishing off-diagonal trace (Lemma 2), the result $G_{\mathrm{eff}}(N) = \sigma(N) \cdot G_{\mathrm{PDL}}$ follows without any additional linearity assumption.

Section 2 recalls the PDL prerequisites, with self-contained tables for condition (B) and the cross-sign enumeration. Section 3 proves the exact independence of gravitational engagements from $(A) \wedge (B)$. Sections 4 and 5 establish Lemmas 1 and 2. Section 6 states and proves Theorem 1. Sections 7 and 8 give numerical checks and open problems.

2 PDL prerequisites

2.1 The K_4 closure, pulsation, and condition (B)

K_4 is the complete signed graph on four vertices $\{0, 1, 2, 3\}$ and six edges $E(K_4) = \{\{i, j\} : i < j\}$, with triangular coherence $s_{ij}^{(1)} s_{jk}^{(1)} s_{ik}^{(1)} = +1$ for every triangle. The binary pulsation alternates between $s^{(1)}$ and $s^{(2)} = -s^{(1)}$ [8]. There are exactly 8 coherent sign configurations of K_4, forming 4 pulsation pairs $\{s, -s\}$.

A mixed triangle (e_i, e_j, p_k) with $e_i, e_j \in V(K_4)$ and p_k a proton vertex is *stable* across both half-cycles if and only if [2]:

$$\text{(A)} \quad P_1 = s^{(1)}(e_i, e_j), \tag{1}$$

$$\text{(B)} \quad P_2 = -P_1, \tag{2}$$

where $P_c = s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c)$ is the cross-product at half-cycle c. This was verified exhaustively over 768 cases with zero counter-example. Table 1 provides a self-contained verification; Table 2 lists all four cross-sign configurations satisfying $(A) \wedge (B)$ for $s^{(1)}(e_i, e_j) = +1$.

Table 1: Self-contained verification of the stability criterion $(A) \wedge (B)$ for mixed triangles. The sign $s^{(1)}(e_i, e_j) = +1$ without loss of generality. Counts over 768 cases (8 coherent K_4 configurations $\times$ 6 edges $\times$ 2^4 cross-sign combinations); 192 stable; 0 counter-examples.

P_1	P_2	(A): $P_1 = s^{(1)}$	(B): $P_2 = -P_1$	$(A) \wedge (B)$	Stable
$+1$	$+1$	✓	×	×	No
$+1$	-1	✓	✓	✓	Yes
-1	$+1$	×	✓	×	No
-1	-1	×	×	×	No

$P_2/P_1 = -1$ in all 192 stable cases. $(A) \wedge (B) \Leftrightarrow$ stable: 768/768.

2.2 Surface coverage fraction and the bridge formula

Definition 2.1 (Surface coverage fraction, D25 [5])**.** The surface coverage fraction of an ensemble of N nucleons is $\sigma(N) = 1 - (1 - \kappa)^N$ with $\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$, where $\varphi = (1 + \sqrt{5})/2$. This satisfies the exact recurrence $\sigma(N + 1) = \sigma(N) + \kappa(1 - \kappa)^N$, verified to machine precision for all $N \geq 0$.

Definition 2.2 (Bridge formula, D20 [3])**.** The gravitational constant is derived as $G_{\mathrm{PDL}} = (\hbar c/m_p^2) \cdot \varepsilon_G^{18}$, where $\varepsilon_G = \Delta r_{\mathrm{val}}/(R \cdot C)$ is the leakage parameter of a single proton, $R =$

Table 2: Exhaustive enumeration of the 4 cross-sign configurations $(s^{(1)}_{\times,i}, s^{(1)}_{\times,j}, s^{(2)}_{\times,i}, s^{(2)}_{\times,j}) \in \{+1,-1\}^4$ satisfying $(A) \wedge (B)$ with $s^{(1)}(e_i, e_j) = +1$. These are the 4 configurations out of 16 for which $P_1 = s^{(1)}_{\times,i} \cdot s^{(1)}_{\times,j} = +1$ and $P_2 = s^{(2)}_{\times,i} \cdot s^{(2)}_{\times,j} = -1$. The fraction $4/16 = 1/4$ is the cross-sign engagement probability κ_{raw} at the level of a single edge.

$s^{(1)}_{\times,i}$	$s^{(1)}_{\times,j}$	$s^{(2)}_{\times,i}$	$s^{(2)}_{\times,j}$	P_1	P_2
-1	-1	-1	$+1$	$+1$	-1
-1	-1	$+1$	-1	$+1$	-1
$+1$	$+1$	-1	$+1$	$+1$	-1
$+1$	$+1$	$+1$	-1	$+1$	-1

4 out of 16 cross-sign combinations satisfy $(A) \wedge (B)$: $\kappa_{\text{raw}} = 4/16 = 1/4$.

$R_{\text{tot}} R_{\text{surf}}/r^2_{\text{val}}$, and $C = (r_{\text{val}} - 1)/r_{\text{val}}$. The saturated bridge amplitude is $\Delta r_{\text{val}}(\infty) = \varepsilon_G \cdot R \cdot C$. Paper D31 proved $\Delta r_{\text{val}}(N) = \sigma(N) \cdot \Delta r_{\text{val}}(\infty)$ to error $< 10^{-12}$ for $N = 0, \ldots, 120$ [6].

2.3 The two hypotheses of Gate 3 (D36)

The proof of Gate 3 in the present paper rests on two explicitly named hypotheses.

Hypothesis 2.3 (H1 — single-proton engagement fraction)**.** $\kappa = R_{\text{surf}}/R_{\text{tot}}$ is the gravitational engagement fraction of a single proton: it is the probability that a given edge $e \in E(K_4)$ belongs to the active gravitational surface of one proton drawn uniformly from the ensemble. This is the interpretation of κ adopted in D25 [5].

Hypothesis 2.4 (H2 — proportionality of G to engagement fraction)**.** The effective gravitational coupling is proportional to the fraction of active gravitational surface: $G_{\text{eff}}(N) = \kappa_{\text{eff}}(N) \cdot G_0$ for some G_0 fixed by the single-proton bridge formula D20 [3].

Remark 2.5 (Relationship to D31)**.** The linear bridge inversion hypothesis of D31 [6] was the statement that G_{eff} scales linearly with $\sigma(N)$, without identifying the mechanism. Hypotheses H1 and H2 of the present paper replace it with a more precise statement: H1 identifies the variable (κ as engagement fraction) and H2 identifies the proportionality (from the bridge). Together they imply the D31 hypothesis as a corollary, while being individually weaker and more transparently motivated. The derivation of H1 from axioms C1–C4 is open problem OP1 (Section 8).

3 Exact independence of gravitational engagements

Proposition 3.1 (Exact independence under $(A) \wedge (B)$)**.** *Let p_k and p_l be two distinct protons ($k \neq l$). For any edge $e = (e_i, e_j) \in E(K_4)$ and any coherent sign configuration $s^{(1)}$ of K_4, the events*

$$A_k = \{p_k \text{ satisfies } (A) \wedge (B) \text{ on } e\}, \qquad A_l = \{p_l \text{ satisfies } (A) \wedge (B) \text{ on } e\} \tag{3}$$

are exactly statistically independent: $P(A_k \cap A_l) = P(A_k) \cdot P(A_l)$.

Proof. The cross-product $P_c(p_k)$ at half-cycle c for proton p_k on edge (e_i, e_j) is $P_c(p_k) = s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c)$. The four cross-signs $(s^{(1)}_{\times,i}, s^{(1)}_{\times,j}, s^{(2)}_{\times,i}, s^{(2)}_{\times,j})_k \in \{+1,-1\}^4$ are the internal variables of proton p_k; the analogous four-tuple for proton p_l is entirely distinct, sharing no variable with that of p_k since $k \neq l$. The condition $(A) \wedge (B)$ for p_k — that $P_1(p_k) = s^{(1)}(e_i, e_j)$ and $P_2(p_k) = -P_1(p_k)$ — depends exclusively on $(s^{(1)}_{\times,i}, s^{(1)}_{\times,j}, s^{(2)}_{\times,i}, s^{(2)}_{\times,j})_k$. Since the cross-sign spaces

of p_k and p_l are disjoint, the events A_k and A_l are defined on independent product spaces, hence independent. Formally:

$$P(A_k \cap A_l) = \frac{|\{(c_k, c_l) \in \{+1,-1\}^4 \times \{+1,-1\}^4 : c_k \in A_k,\, c_l \in A_l\}|}{16^2}$$
$$= \frac{|A_k|}{16} \cdot \frac{|A_l|}{16} = P(A_k) \cdot P(A_l). \tag{4}$$

From Table 2: $|A_k| = |A_l| = 4$, so $P(A_k) = P(A_l) = 4/16 = 1/4$. The joint count is 16 out of 256, giving $P(A_k \cap A_l) = 1/16 = (1/4)^2$. The correlation is $\delta = 1/16 - (1/4)^2 = 0$ exactly. □

Table 3 provides the exhaustive numerical verification of Proposition 3.1.

Table 3: Exhaustive verification of exact independence: $P(A_k \cap A_l) = P(A_k) \cdot P(A_l) = (1/4)^2$. All $256 = 16^2$ cross-sign pairs are enumerated. The correlation $\delta = P(\text{joint}) - P(\text{marginal})^2$ is exactly zero.

Quantity	Value	Comment
$P(A_k) = \lvert A_k\rvert/16$	$4/16 = 0.2500$	cross-sign level
$P(A_l) = \lvert A_l\rvert/16$	$4/16 = 0.2500$	cross-sign level
$P(A_k \cap A_l)$	$16/256 = 0.0625$	exhaustive enumeration
$P(A_k) \cdot P(A_l)$	$0.2500^2 = 0.0625$	independence prediction
$\delta = P(A_k \cap A_l) - P(A_k) \cdot P(A_l)$	0 (exact)	
$\kappa = R_{\text{surf}}/R_{\text{tot}}$	0.045529	PDL engagement fraction
$P(A \cap B)$ at κ level	$\kappa^2 = 0.002073$	
$2\kappa - \sigma(2)$	0.002073	$= \kappa^2$ algebraically
$\delta_{\text{PDL}} = \kappa^2 - \kappa^2$	0 (exact)	

Algebraic proof: $\sigma(2) = 2\kappa - \kappa^2$, so $P(A \cap B) = 2\kappa - \sigma(2) = \kappa^2 = P(A) \cdot P(B)$. □

Corollary 3.2 (Independence at the κ level)**.** *At the level of the surface engagement fraction $\kappa = R_{\text{surf}}/R_{\text{tot}}$, the independence identity $P(A_k \cap A_l) = \kappa^2$ follows algebraically from the definition of $\sigma(N)$:*

$$\sigma(2) = 1 - (1-\kappa)^2 = 2\kappa - \kappa^2, \quad P(A_k \cap A_l) = 2\kappa - \sigma(2) = \kappa^2 = \kappa \cdot \kappa. \tag{5}$$

The correlation at the κ level is therefore $\delta_{\text{PDL}} = \kappa^2 - \kappa \cdot \kappa = 0$ exactly, for any value of κ.

Remark 3.3 (Independence is contained in $\sigma(N)$, not an additional assumption)**.** Corollary 3.2 shows that the independence of gravitational engagements is not a separate hypothesis: it is algebraically equivalent to the definition $\sigma(N) = 1 - (1-\kappa)^N$, which is the union probability of N independent Bernoulli events of probability κ. The formula is self-consistent: $\sigma(N)$ is correct if and only if the engagements are independent, and the independence follows from the formula. The residual open question is therefore not about independence but about the value of κ itself (OP1, Section 8).

4 Uniform occupation and the active-surface projector

Lemma 4.1 (Uniform occupation, Lemma 1)**.** *Under Hypothesis H1 and Proposition 3.1, for any edge $e \in E(K_4)$ and any $N \geq 0$:*

$$P(e \text{ active} \mid N \text{ protons}) = \sigma(N) = 1 - (1-\kappa)^N. \tag{6}$$

This holds uniformly for all six edges of K_4.

Proof. By Hypothesis H1, each proton activates edge e independently with probability κ. By Proposition 3.1, the activation events of distinct protons on the same edge are exactly independent. The event that edge e is active (activated by at least one of the N protons) has probability $1 - P(\text{all } N \text{ protons inactive on } e) = 1 - (1-\kappa)^N = \sigma(N)$ by the inclusion-exclusion principle for independent events. The uniformity over all six edges of K_4 follows from the S_4-transitivity proved in D31 [6]: S_4 acts transitively on $E(K_4)$, placing all six edges in a single orbit of size 6, so every S_4-invariant quantity takes the same value on each edge. Under Hypothesis H1, κ is S_4-invariant (it is a scalar), so the activation probability κ is the same for every edge. □

Remark 4.2 (Active-surface projector)**.** Lemma 4.1 implies that the expected active-surface projector is $\mathbb{E}[P_{\text{active}}(N)] = \sigma(N) \cdot I_6$, where I_6 is the identity on $\mathbb{R}^6$ (the edge space of K_4). This is a diagonal projector with all diagonal entries equal to $\sigma(N)$: it assigns the same occupation probability to every edge, a consequence of both independence (Proposition 3.1) and S_4-uniformity (Lemma 4.1).

5 Trace cancellation and the coupling matrix

Lemma 5.1 (Trace cancellation, Lemma 2)**.** *Let A be the edge-adjacency matrix of K_4, defined by $A_{ij} = 1$ if edges i and j share a vertex of K_4, and $A_{ij} = 0$ otherwise, with $A_{ii} = 0$. Then* $\mathrm{Tr}(A) = 0$.

Proof. $\mathrm{Tr}(A) = \sum_{i=1}^{6} A_{ii} = 0$ by definition, since $A_{ii} = 0$: no edge is adjacent to itself. □

Proposition 5.2 (β does not contribute to $G_{\text{eff}}(N)$)**.** *The most general S_4-invariant coupling matrix on $\mathbb{R}^6$ has the form $\Phi = \alpha I_6 + \beta A$ for $\alpha, \beta \in \mathbb{R}$. For any such Φ and for the active-surface projector $P_{\text{active}}(N) = \sigma(N) \cdot I_6$ of Remark 4.2:*

$$\mathrm{Tr}(\Phi \cdot P_{\text{active}}(N)) = \alpha \cdot \mathrm{Tr}(I_6) \cdot \sigma(N) + \beta \cdot \mathrm{Tr}(A) \cdot \sigma(N) = 6\alpha \cdot \sigma(N). \tag{7}$$

The term in β vanishes identically for all N and all β by Lemma 5.1.

Proof. Since $P_{\text{active}}(N) = \sigma(N) \cdot I_6$:

$$\begin{aligned} \mathrm{Tr}(\Phi \cdot P_{\text{active}}(N)) &= \sigma(N) \cdot \mathrm{Tr}(\Phi) = \sigma(N) \cdot (\alpha \cdot \mathrm{Tr}(I_6) + \beta \cdot \mathrm{Tr}(A)) \\ &= \sigma(N) \cdot (6\alpha + \beta \cdot 0) = 6\alpha \cdot \sigma(N). \end{aligned} \tag{8}$$

The S_4-invariance of $\Phi = \alpha I_6 + \beta A$ is verified by direct computation: $[P_\pi, I_6] = 0$ and $[P_\pi, A] = 0$ for all 24 elements $\pi \in S_4$, where P_π is the permutation matrix on $\mathbb{R}^6$ induced by the action of π on $E(K_4)$. This was verified numerically to error $< 10^{-14}$ for all 24 elements (Section 7). □

Remark 5.3 (Structural meaning of trace cancellation)**.** Proposition 5.2 shows that whether or not adjacent edge-pairs are correlated in the coupling (i.e. regardless of β), the gravitational coupling is determined entirely by the diagonal part αI_6 of Φ. The structure of K_4 — specifically the fact that no edge is self-adjacent — renders the off-diagonal part inert. This is a combinatorial property of K_4, valid for any graph, and does not depend on any physical assumption.

6 Gate 3 without linear bridge inversion hypothesis

Theorem 6.1 (Gate 3 strengthened)**.** *Under Hypothesis H1 (Definition 2.3) and Hypothesis H2 (Definition 2.4), for all $N \geq 0$:*

$$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}} = \left[1 - (1-\kappa)^N\right] \cdot G_{\text{PDL}}. \tag{9}$$

The linear bridge inversion hypothesis of D31 [6] is superseded: linearity in $\sigma(N)$ is a consequence of H1, H2, Proposition 3.1, and Lemma 4.1, not an independent assumption.

Proof. By Hypothesis H2, $G_{\text{eff}}(N) = \text{Tr}(\Phi \cdot P_{\text{active}}(N)) \cdot (\text{rank}(d_2)/n_e)$ where $n_e = 6$ is the number of edges of K_4 and $\text{rank}(d_2) = 4$ is the rank of the gravitational-level transition map of the PDL chain complex [11, 6]. By Proposition 5.2, $\text{Tr}(\Phi \cdot P_{\text{active}}(N)) = 6\alpha \cdot \sigma(N)$ for any S_4-invariant Φ. By the saturated normalisation condition $G_{\text{eff}}(\infty) = G_{\text{PDL}}$: $6\alpha \cdot 1 \cdot (4/6) = G_{\text{PDL}}$, giving $\alpha = G_{\text{PDL}}/4$. Substituting:

$$G_{\text{eff}}(N) = 6 \cdot \frac{G_{\text{PDL}}}{4} \cdot \sigma(N) \cdot \frac{4}{6} = \sigma(N) \cdot G_{\text{PDL}}. \tag{10}$$

The value $\alpha = G_{\text{PDL}}/4$ is consistent with the bridge formula of D20 (Hypothesis H2): $G_{\text{PDL}} = (\hbar c/m_p^2) \cdot \varepsilon_G^{18}$ at saturation, so $\alpha = (\hbar c/m_p^2) \cdot \varepsilon_G^{18}/4$ is the contribution of a single active edge to the gravitational coupling. At $N = 0$: $\sigma(0) = 0$ and $G_{\text{eff}}(0) = 0$ (no active surface). At $N \to \infty$: $\sigma(N) \to 1$ and $G_{\text{eff}} \to G_{\text{PDL}}$. □

Remark 6.2 (Why $\sigma(N)^{18} \cdot G_{\text{PDL}}$ is ruled out)**.** A naive substitution $\varepsilon_G \mapsto \sigma(N) \cdot \varepsilon_G$ in the bridge formula $G_{\text{PDL}} = (\hbar c/m_p^2) \cdot \varepsilon_G^{18}$ would give $G_{\text{eff}}(N) = \sigma(N)^{18} \cdot G_{\text{PDL}}$. This is incompatible with the verified Gate 3 result: at $N = 40$, $\sigma(40) = 0.8449$ while $\sigma(40)^{18} = 0.0482$, a factor of 17.5 discrepancy. The resolution is that ε_G^{18} is an intrinsic property of the single-proton chain complex, and it must not be replaced by $(\sigma(N) \cdot \varepsilon_G)^{18}$. The correct variable is not $\varepsilon_G(N)$ but the active-surface projector $P_{\text{active}}(N)$, whose trace gives $\sigma(N)$ linearly by Lemma 4.1.

Remark 6.3 (Epistemic advance over D31)**.** The proof of Theorem 6.1 replaces the single unnamed hypothesis of D31 with the following explicit chain: H1 (named, from D25) ⇒ Proposition 3.1 (proved from cross-sign disjointness) ⇒ Lemma 4.1 (proved from independence + S_4-transitivity) ⇒ Proposition 5.2 (proved from $\text{Tr}(A) = 0$) ⇒ Theorem 6.1 (proved from H2 + normalisation). The sole residual open problem is the derivation of H1 from axioms C1–C4 (OP1, Section 8).

7 Numerical checks

All numerical checks were performed in Python 3 (NumPy 2.x). Python code is available from the author upon request.

Table 4: Numerical verification of the main results of D36.

Quantity	Expression verified	Max. residual
$(A) \wedge (B)$ exhaustive count	4/16 configurations	exact
Independence δ at cross-sign level	$16/256 - (4/16)^2$	0 (exact)
Independence δ_{PDL} at κ level	$\kappa^2 - \kappa \cdot \kappa$	7.6×10^{-17}
$\text{Tr}(A) = 0$	sum of diagonal of A	0 (exact)
S_4-invariance of I_6: $\max\lvert P_\pi I_6 - I_6 P_\pi\rvert$	all 24 elements	0.00×10^0
S_4-invariance of A: $\max\lvert P_\pi A - A P_\pi\rvert$	all 24 elements	0.00×10^0
$\text{Tr}(\Phi \cdot P_{\text{active}})$ for $\beta = 5\alpha$	β term $= 0$	0.00×10^0
$\sigma(N+1) = \sigma(N) + \kappa(1-\kappa)^N$	$N = 0, \ldots, 120$	1.1×10^{-16}
$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ vs bridge (D20)	$N = 0, \ldots, 200$	27 ppm
Hubble ratio $\sqrt{\sigma(120)/\sigma(40)}$	PDL vs obs. 1.085935	0.006%

8 Open problems

OP1 (Derivation of H1 from axioms C1–C4). Hypothesis H1 states that $\kappa = R_{\text{surf}}/R_{\text{tot}}$ is the gravitational engagement fraction of a single proton. It is adopted as the interpretation of κ in D25 [5], where $R_{\text{surf}} = 310\varphi$ is the active surface budget and $R_{\text{tot}} = 11017$ is the total relational

Table 5: Sensitivity of the Hubble tension ratio to residual correlations δ between proton engagements. A correlation $\delta = 0.001$ degrades the agreement by 0.3526%, which is 59× the current 0.006% precision. The independence assumption is therefore precise to $\delta < 0.0001$ relative to the observational constraint.

δ	Effective κ_{eff}	Hubble ratio	Residual from obs.
0	0.045529	1.085868	0.006%
0.0001	0.045629	1.085485	0.035%
0.001	0.046529	1.082039	0.353%
0.005	0.050529	1.068400	1.609%
0.010	0.055529	1.054565	2.883%

Observed ratio: $H_{0,\text{loc}}/H_{0,\text{CMB}} = 73.04/67.26 = 1.085935$ [12, 10].

budget of the proton. The open problem is to derive this identification from axioms C1–C4 alone, without invoking D25 as a definition. Concretely: showing that the four PDL axioms force the ratio $R_{\text{surf}}/R_{\text{tot}}$ to equal the probability that a single proton–electron mixed triangle is in a gravitationally active configuration.

OP2 (Exclusion of G_{sea} from stable engagement). Paper D29 showed that the sea contribution $(1/4)^N \to 0$ probabilistically [2]. A strict algebraic proof that the PDL pulsation axiom structurally excludes G_{sea} from stable gravitational engagement would strengthen the derivation of the occupation probability at the cross-sign level ($\kappa_{\text{raw}} = 1/4$ from Table 2).

OP3 (Structural derivation of N_{loc}). The Hubble tension prediction requires $N_{\text{loc}} = 120$ to be inferred from the observed local H_0. The value $N_{\text{CMB}} = 40$ is derived from the neutron mass gap [10]; an analogous structural derivation of N_{loc} from a PDL model of cosmological structure formation would make the prediction fully parameter-free.

Conclusion

This paper has replaced the linear bridge inversion hypothesis of D31 with an explicit proof chain. The main structural result is that the linearity of $G_{\text{eff}}(N)$ in $\sigma(N)$ is a consequence of three independent facts: (i) the gravitational engagements of distinct protons are exactly independent (Proposition 3.1), because their cross-sign spaces are disjoint under $(A) \wedge (B)$; (ii) the active-surface projector is therefore $P_{\text{active}}(N) = \sigma(N) \cdot I_6$ by S_4-transitivity (Lemma 4.1); and (iii) the edge-adjacency matrix A has $\text{Tr}(A) = 0$, so the off-diagonal part βA of any S_4-invariant coupling matrix contributes nothing to $G_{\text{eff}}(N)$ for any value of β (Lemma 5.1, Proposition 5.2).

The naive substitution $\varepsilon_G \mapsto \sigma(N) \cdot \varepsilon_G$ in the bridge formula would yield $\sigma(N)^{18} \cdot G_{\text{PDL}}$, which is incompatible with Gate 3 by a factor of 17.5 at $N = 40$ (Remark 6.2). This diagnostic confirms that the power-18 structure is intrinsic to a single proton, not to the collective density. The correct collective variable is $P_{\text{active}}(N)$, not $\varepsilon_G(N)$.

Gate 3 is proved under H1 and H2 (Theorem 6.1). The sole residual open problem is the derivation of H1 from axioms C1–C4 (OP1): why is $\kappa = R_{\text{surf}}/R_{\text{tot}}$ the gravitational engagement probability of a single proton? This question is more precise and more tractable than the original open problem of D31, and its resolution would make Gate 3 fully unconditional.

References

[1] CODATA. Codata internationally recommended 2022 values of the fundamental physical constants. https://physics.nist.gov/cuu/Constants/, 2022. $G_N = 6.67430 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$.

[2] C. Laubscher. Axiomatic derivation of the leading valence engagement amplitude in the PDL framework. *Zenodo*, 2026. PDL corpus, D29. Gate 1 resolved: 155/11017 from PDL axioms. Conditions (A) and (B) for mixed-triangle stability; cross-product $P_c = s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c)$; 768 cases, 192 stable, 0 counter-examples; $(1/4)^N \to 0$ probabilistic exclusion of G_{sea}.

[3] C. Laubscher. Bridge formula $\delta r_{\text{val}} = \varepsilon_g \cdot r \cdot c$ and the gravitational constant in the PDL framework. *Zenodo*, 2026. PDL corpus, D20-V3. $G_{\text{PDL}} = (\hbar c/m_p^2) \cdot \varepsilon_G^{18} = 6.67448 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$, deviation 27 ppm from CODATA 2022; $\varepsilon_G = 0.0075194$; $R = R_{\text{tot}} R_{\text{surf}}/r_{\text{val}}^2$; $C = (r_{\text{val}} - 1)/r_{\text{val}}$.

[4] C. Laubscher. Combinatorial derivation of the proton quintuplet and the fine-structure constant in the PDL framework. *Zenodo*, 2026. PDL corpus, D24. Quintuplet $(24, 28, 930, 10087, 11017)$ unique; $\alpha_{\text{PDL}}^{-1} = \varphi^2 r_{\text{val}}^2/(9\mu) = 137.022$; $R_{\text{surf}} = 310\varphi$; $\mu = R_{\text{tot}}/R_e = 11017/6$.

[5] C. Laubscher. Effective gravitational coupling $g_{\text{eff}}(n) = \sigma(n) \cdot g_{\text{PDL}}$ from closure density in the PDL framework. *Zenodo*, 2026. PDL corpus, D25. $\sigma(N) = 1 - (1-\kappa)^N$ proved from recurrence; $\kappa = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; κ defined as gravitational engagement fraction of a single proton; $N_{1/2} = 14.9$.

[6] C. Laubscher. $g_{\text{eff}}(n) = \sigma(n) \cdot g_{\text{PDL}}$: Gate 3 from the linear bridge inversion hypothesis. *Zenodo*, 2026. PDL corpus, D31. Gate 3 under linear bridge inversion hypothesis; S_4 transitivity on K_4: single orbit of size 6, exhaustive over all 24 elements; d_2^{sat} not S_4-equivariant (physical obstruction); $\Delta r_{\text{val}}(N) = \sigma(N) \cdot \Delta r_{\text{val}}(\infty)$, error $< 10^{-12}$; OP1 = algebraic proof of linearity.

[7] C. Laubscher. Introduction to the projective dynamic logo framework. *Zenodo*, 2026. PDL corpus, D2.

[8] C. Laubscher. The projective dynamic logo: Physical reality from relational axioms on finite signed graphs. *Zenodo*, 2026. PDL corpus, D1. Four axioms C1–C4; K_4 as unique minimal closure; $R_e = 6$; triangular coherence; binary pulsation $s^{(1)} \leftrightarrow s^{(2)} = -s^{(1)}$.

[9] C. Laubscher. Quantitative einstein equation from $g_{\text{eff}}(n) = \sigma(n) \cdot g_{\text{PDL}}$ in the PDL framework. *Zenodo*, 2026. PDL corpus, D35. PDL Einstein coupling $\kappa_{\text{PDL}}(N) = 8\pi\sigma(N) G_{\text{PDL}}/c^4$; three epistemic layers; Hubble tension at 0.006%.

[10] C. Laubscher. Structural derivation of $n_{\text{cmb}} = 40$ from the neutron mass gap and parameter-free hubble constant in the PDL framework. *Zenodo*, 2026. PDL corpus, D27. $N_{\text{CMB}} = \lfloor \Gamma_n \rfloor = 40$ from $\Gamma_n = 6\mu_n - R_{\text{tot}}(n) = 40.102$, no cosmological input; $H_{0,\text{CMB}} = 67.26\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ at 0.27σ Planck.

[11] C. Laubscher. Topological origin of the exponent 18 in the PDL framework: $18 = 6+5+4+3$ and $k_4 \cong \partial\delta^3 \cong s^2$. *Zenodo*, 2026. PDL corpus, D23. PDL chain complex $C_0 \to C_1 \to C_2 \to C_3$ with ranks 6, 5, 4, 3; $\text{rank}(d_2) = 4$ is the rank of the gravitational-level transition map; $\ker(d_2)$ two-dimensional (confined valence modes).

[12] A. G. Riess et al. A comprehensive measurement of the local value of the hubble constant with $1\,\text{km}\,\text{s}^{-1}\,\text{mpc}^{-1}$ uncertainty from the Hubble Space Telescope and the SH0ES team. *Astrophysical Journal Letters*, 934:L7, 2022.